

MrFlexySMPs Folded Horn Hybrid

Build plan · Rev. D · 2026-10-17

Print on A3 at 100 % (no fit to page). All dimensions in mm.

Material: 18 mm multiplex plywood throughout

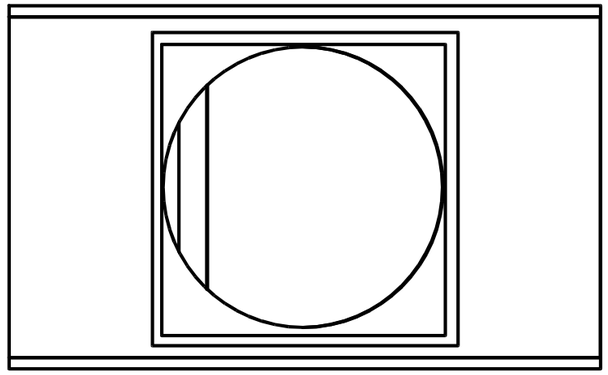
Parts list and cut list: sheet 26

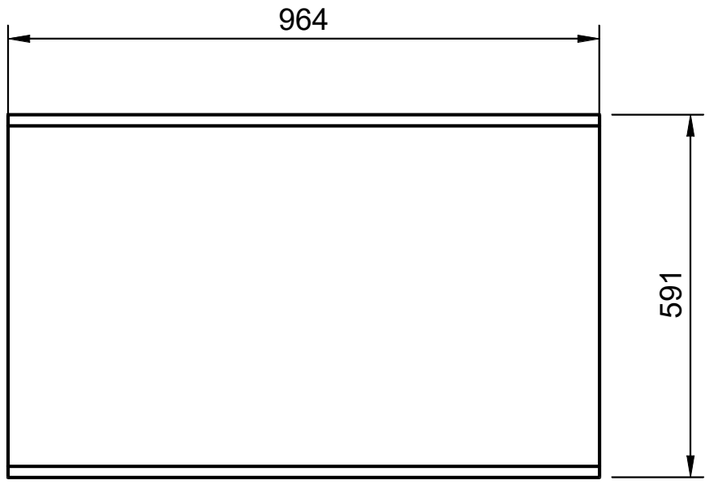
Design: MrFlexySMP, YouTube: https://www.youtube.com/watch?v=tzwWSL8TyKI

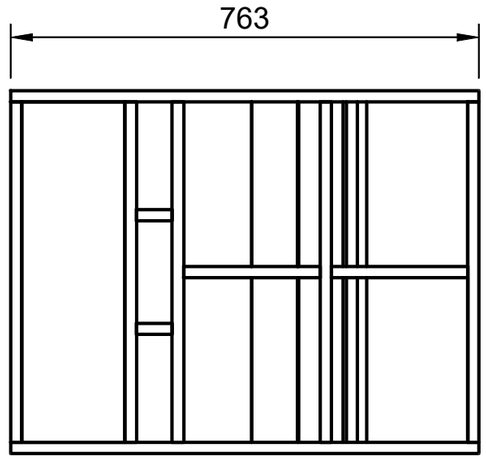
3D model (3dviewer.net):

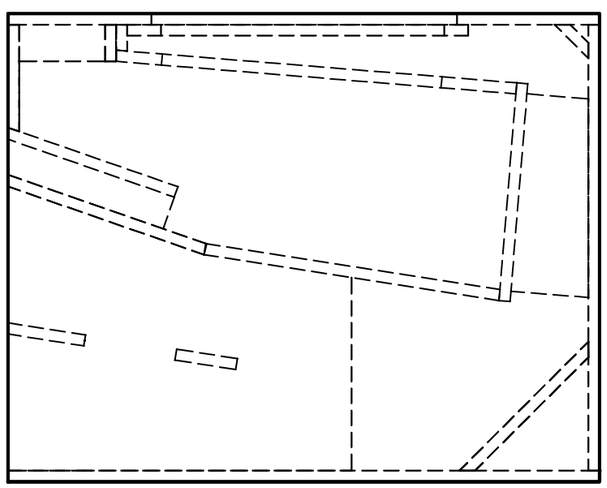
Model files (GitHub):

3D model (3dviewer.net):	https://3dviewer.net/#model=https://raw.githubusercontent.com/SdWa5/3d/main/published/flexy-folded-horn-hybrid/flexy-folded-horn-hybrid.glb
Model files (GitHub):	https://github.com/SdWa5/3d/tree/main/published/flexy-folded-horn-hybrid
Original CAD (v28):	https://3dviewer.net/#model=https://github.com/Oletarier/bassw-rfel-folded-horn/blob/main/subwoofer%20v28.obj









Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:10 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

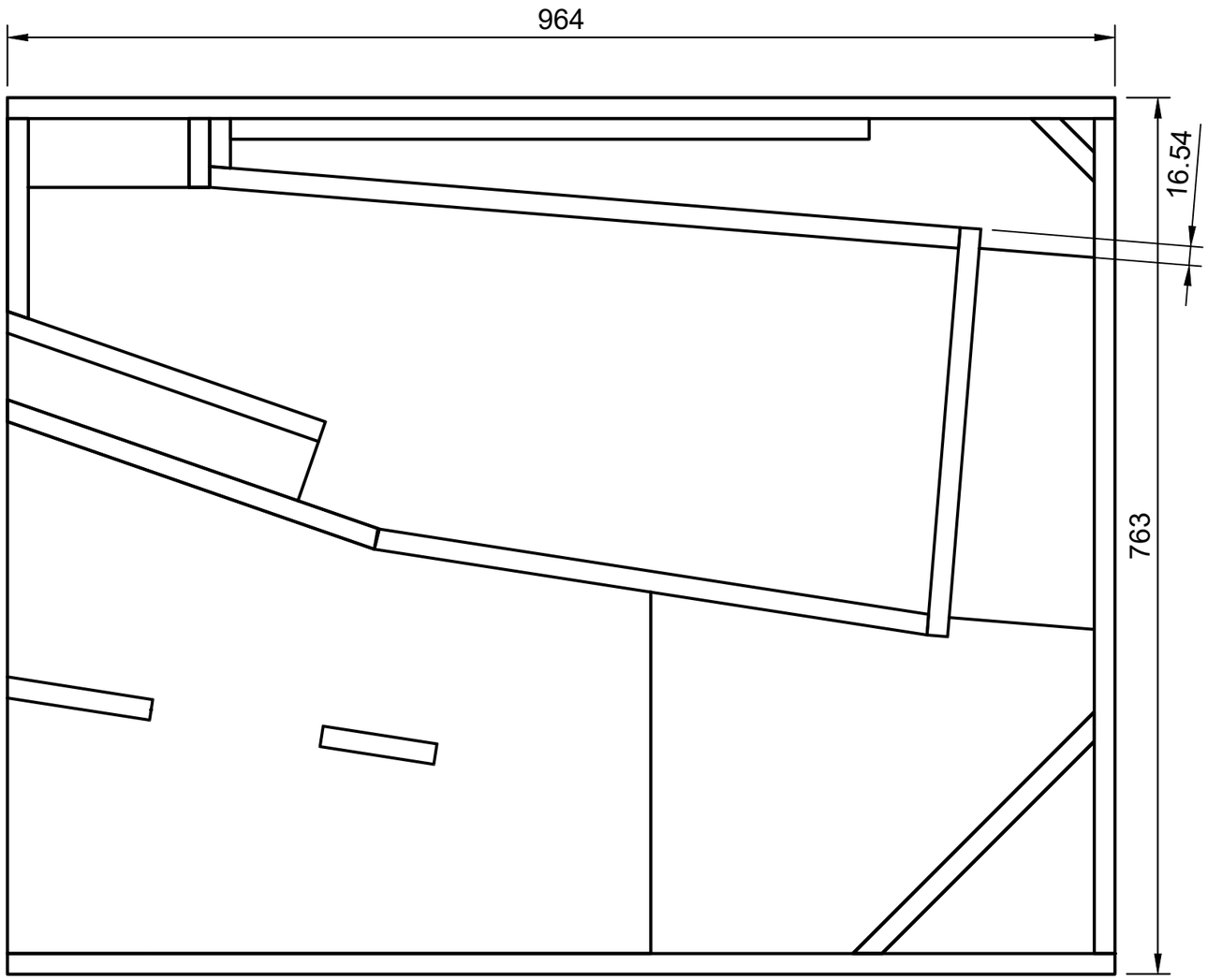
MrFlexySMPs
Folded Horn Hybrid

Assembly views

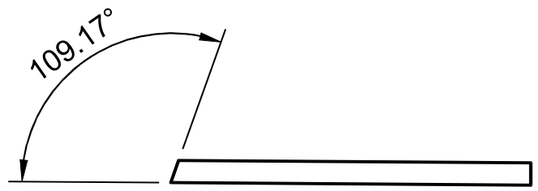
DWG No.

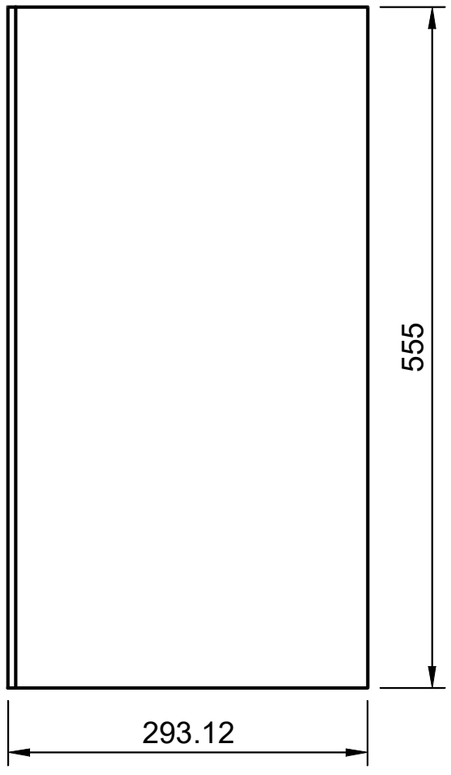
FLEXY-02

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Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper
Material		Document type	Document status
18 mm		Build plan	revised
multiplex plywood		Title MrFlexySMPs Folded Horn Hybrid **Side section**	DWG No. FLEXY-03
			Rev. D / Date of issue 2026-10-17 / Sheet 3/26





Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

MrFlexySMPs
Folded Horn Hybrid

Panel 555 × 293.12

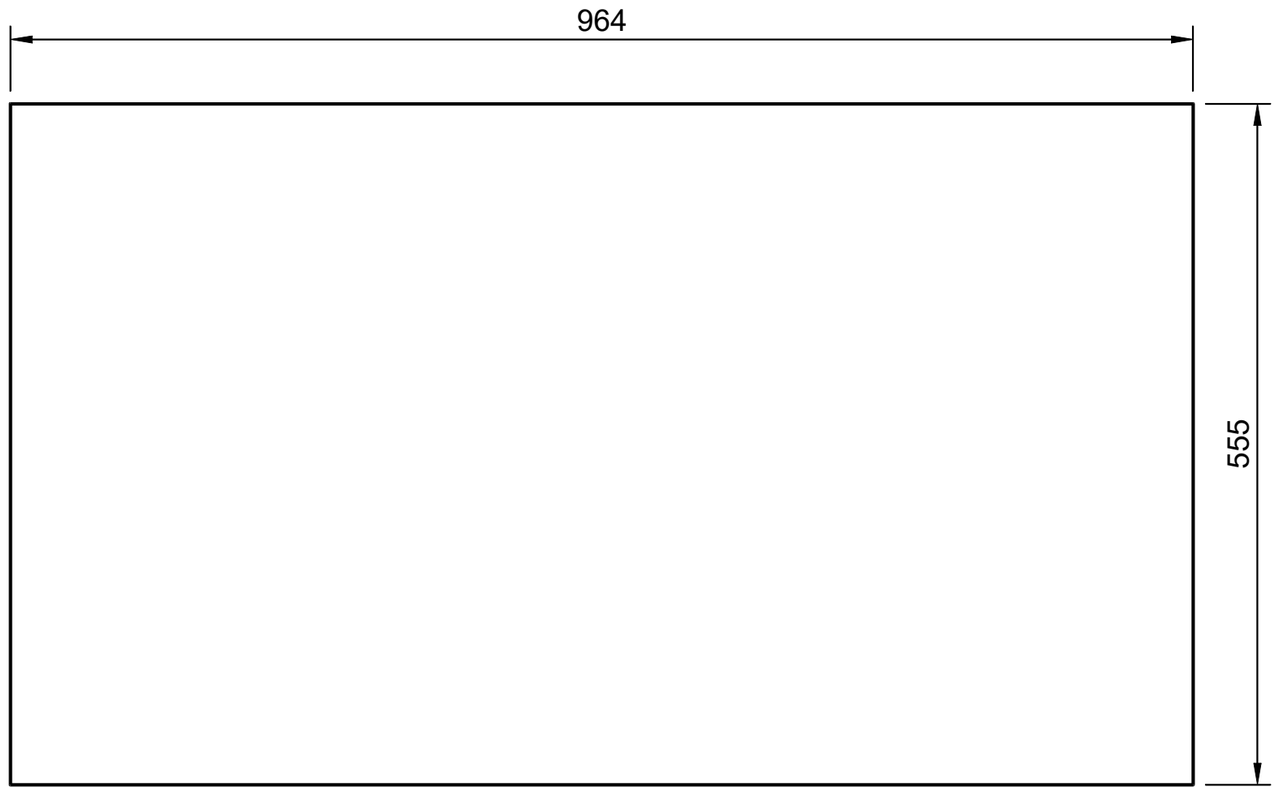
DWG No.

FLEXY-04

Rev.	Date of issue	Sheet
D	2026-10-17	4/26

NOTE

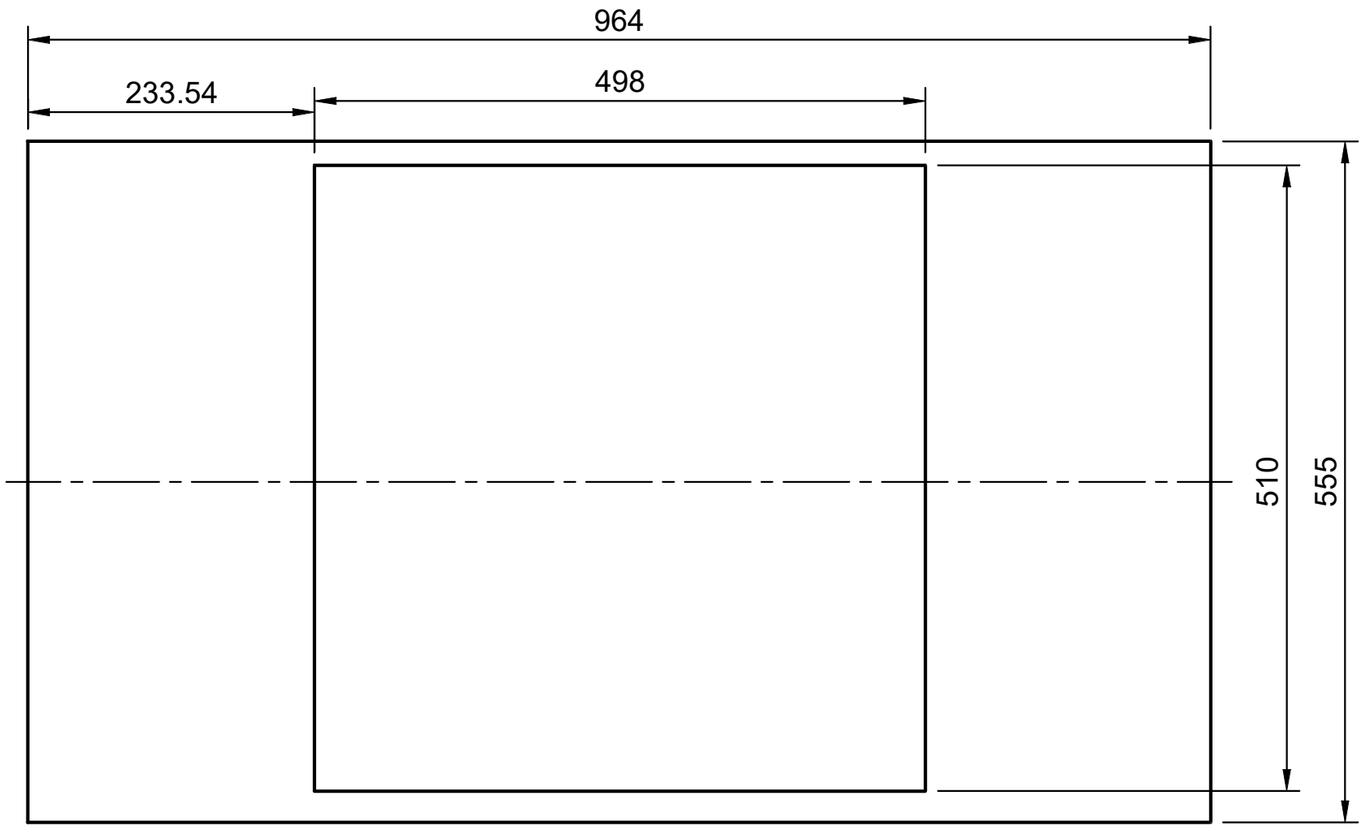
Same blank as the top panel (sheet 6).



Dept. SdWa5	Technical reference Scale 1:5 on A3	Created by Oliver Ripper 25.02.2022	Approved by Stefan Ripper
Material **18 mm** **multiplex plywood**		Document type Build plan	Document status revised
		Title MrFlexySMPs Folded Horn Hybrid **Bottom panel 964 × 555**	DWG No. FLEXY-05
			Rev. D / Date of issue 2026-10-17 / Sheet 5/26

NOTE

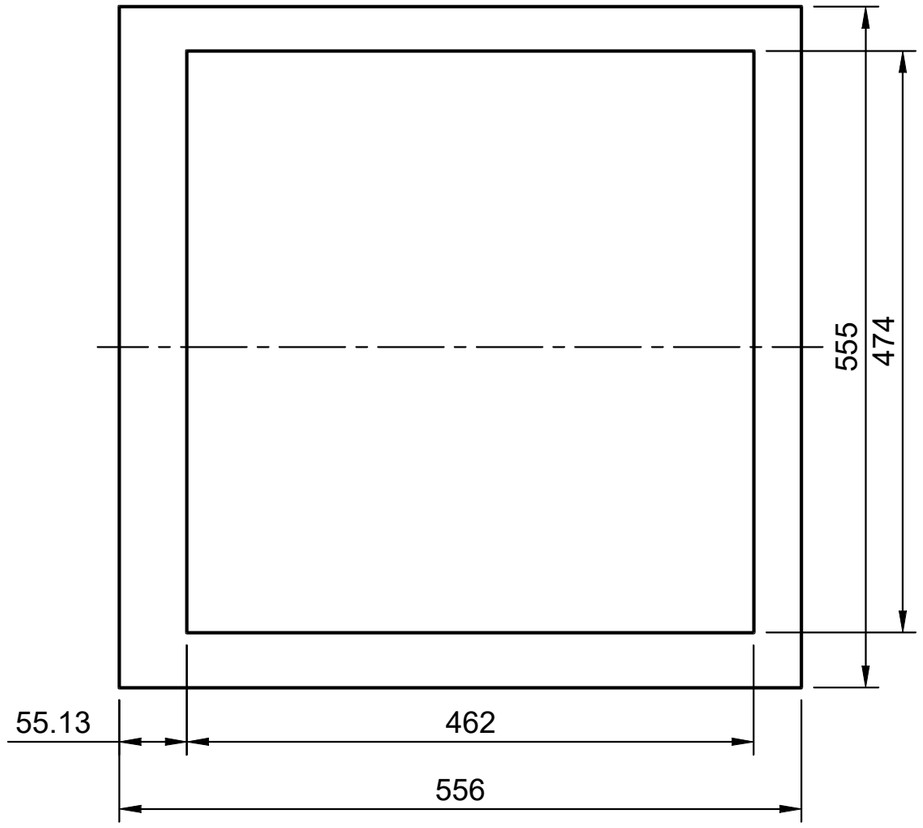
The cut-out piece becomes the lid (sheet 14).



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper
Material: **18 mm** multiplex plywood		Document type: Build plan	Document status: revised
		Title: MrFlexySMPs Folded Horn Hybrid — **Top panel with lid opening**	DWG No.: FLEXY-06
			Rev.: D · Date of issue: 2026-10-17 · Sheet: 6/26

OPTIONAL

The frame may also be built from four separate battens (offcuts).



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

MrFlexySMPs
Folded Horn Hybrid

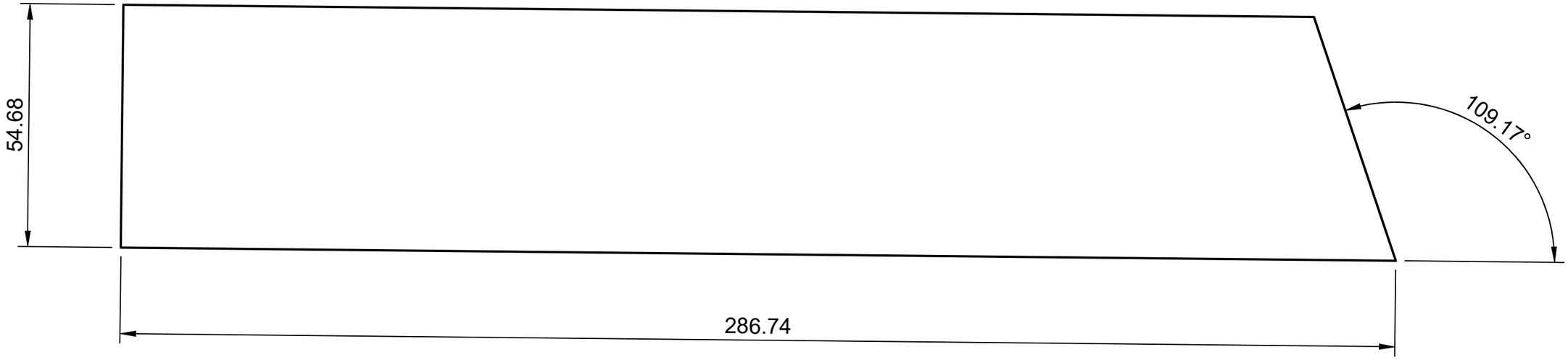
Frame 556 × 555

DWG No.

FLEXY-07

Rev.	Date of issue	Sheet
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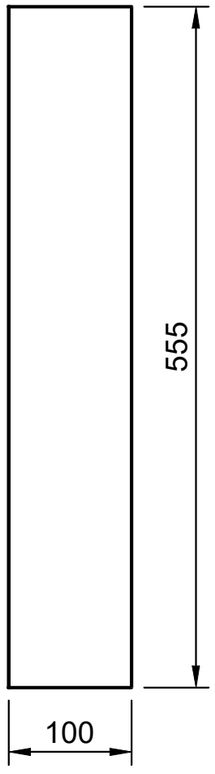
x2



Dept. SdWa5	Technical reference Scale 1:1 on A3	Created by Oliver Ripper 25.02.2022	Approved by Stefan Ripper
Material **18 mm** **multiplex plywood**		Document type Build plan Title MrFlexySMPs Folded Horn Hybrid **Trapezoid strip 286.74 (2×)**	Document status revised DWG No. FLEXY-08
			Rev. D / Date of issue 2026-10-17 / Sheet 8/26

OPTIONAL

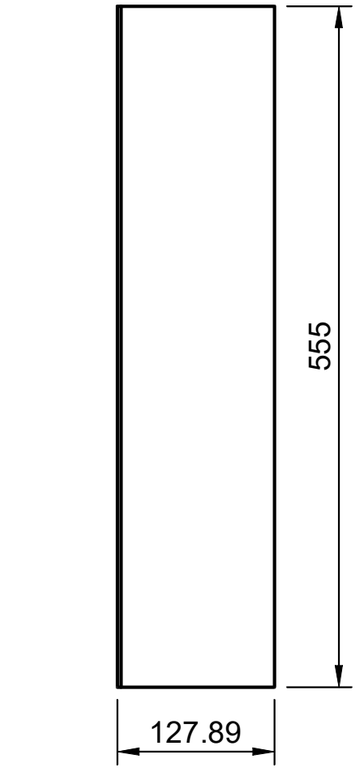
Braces: may also be made of metal, for example.

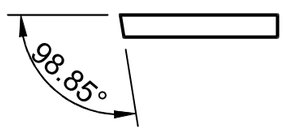


Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper
Material: **18 mm** multiplex plywood		Document type: Build plan	Document status: revised
		Title: MrFlexySMPs Folded Horn Hybrid — **Brace 100 × 555**	DWG No.: FLEXY-09
			Rev.: D · Date of issue: 2026-10-17 · Sheet: 9/26

OPTIONAL

Braces: may also be made of metal, for example.





Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

Document status: revised

Title: MrFlexySMPs
Folded Horn Hybrid

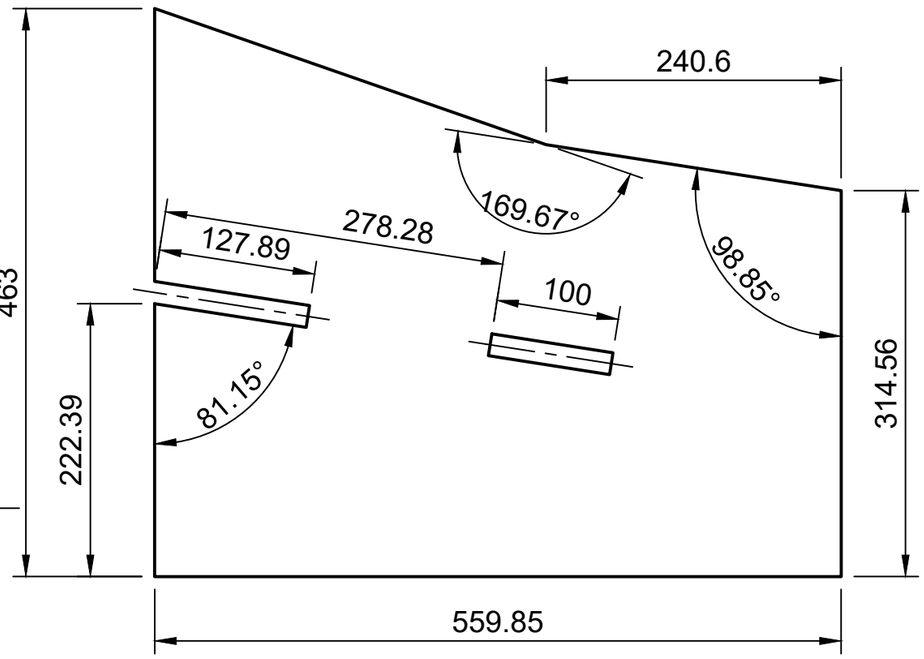
Brace 127.89 × 555

DWG No.: FLEXY-10

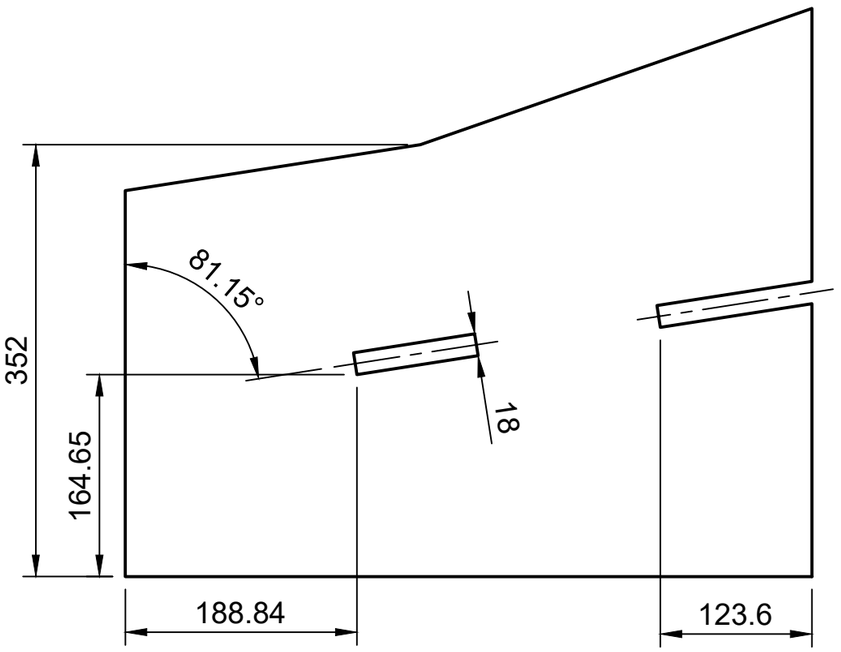
Rev.	Date of issue	Sheet
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OPTIONAL

Braces: may also be made of metal, for example.







Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

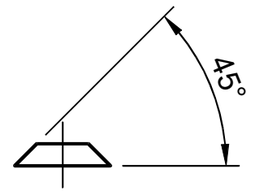
Document status: revised

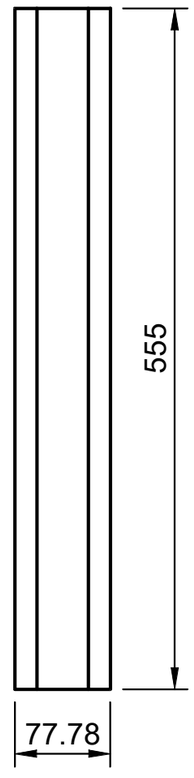
Title: MrFlexySMPs Folded Horn Hybrid

Brace plate

DWG No.: FLEXY-11

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Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

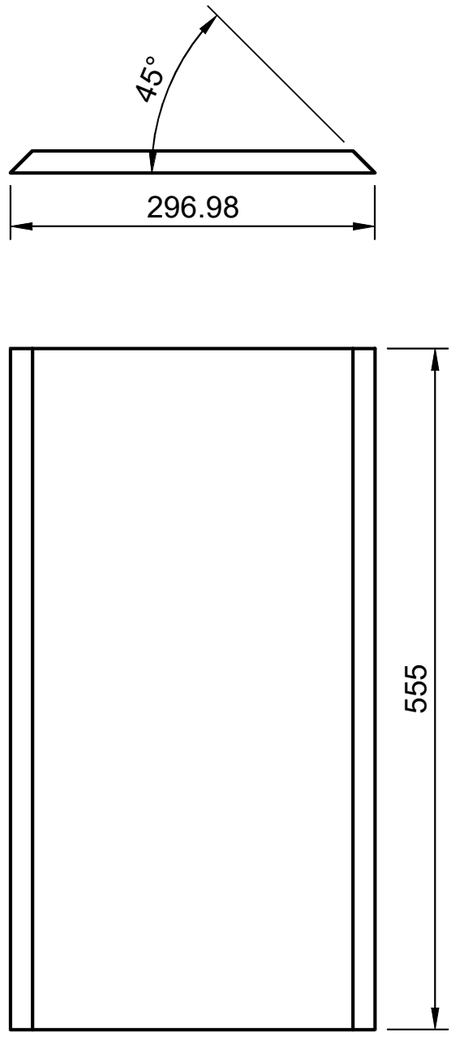
MrFlexySMPs
Folded Horn Hybrid

Strip 77.78 × 555

DWG No.

FLEXY-12

Rev.	Date of issue	Sheet
D	2026-10-17	12/26



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

MrFlexySMPs
Folded Horn Hybrid

Panel 296.98 × 555

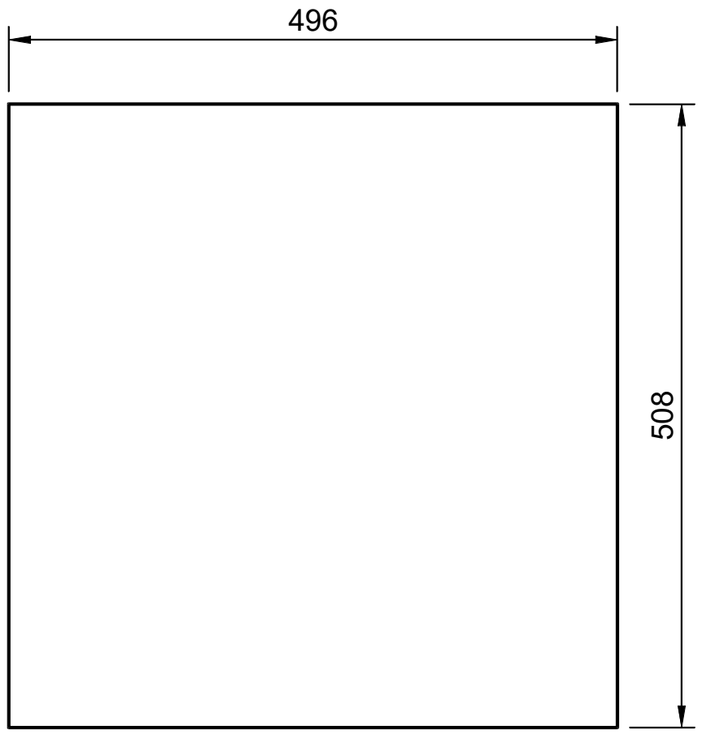
DWG No.

FLEXY-13

Rev.	Date of issue	Sheet
D	2026-10-17	13/26

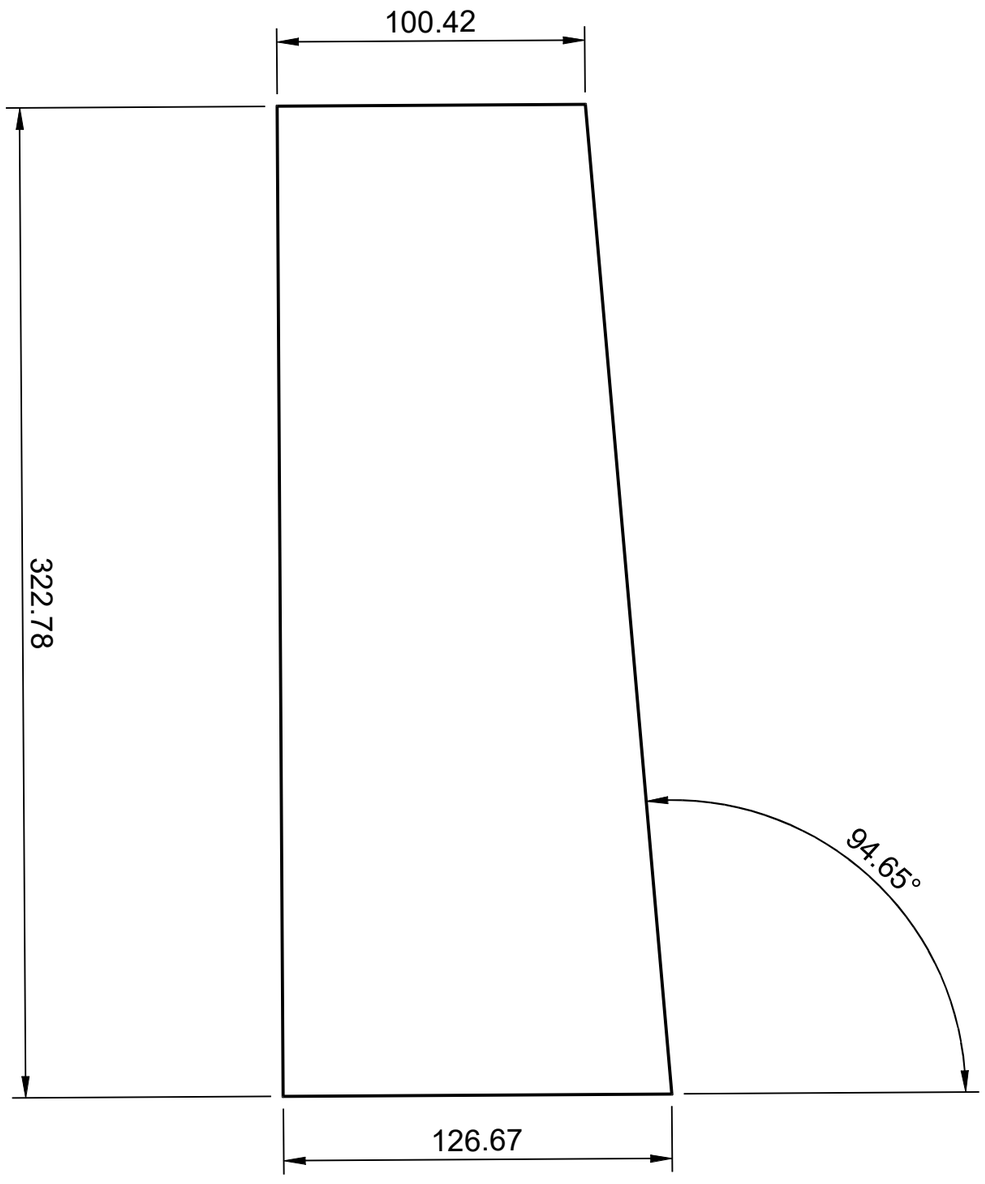
OPTIONAL

The lids are cut out of the top panel (sheet 6).



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper
Material: **18 mm** multiplex plywood		Document type: Build plan	Document status: revised
		Title: MrFlexySMPs Folded Horn Hybrid **Lid 496 × 508**	DWG No.: FLEXY-14
			Rev. D · Date of issue 2026-10-17 · Sheet 14/26

x2



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:2 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

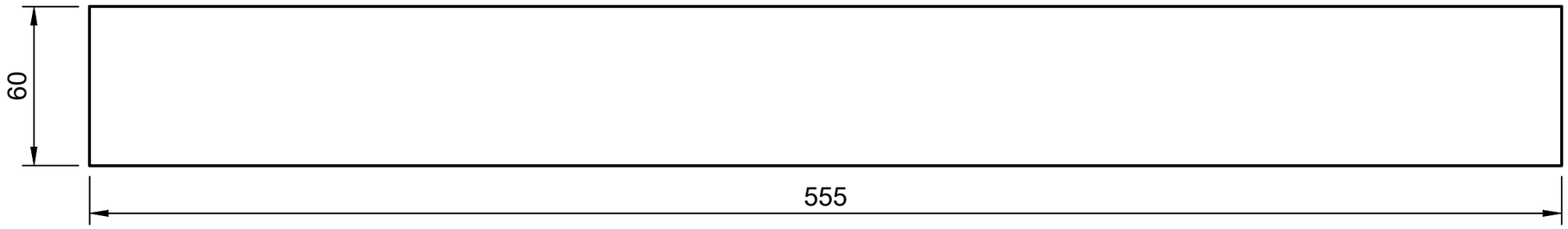
MrFlexySMPs
Folded Horn Hybrid

Trapezoid 322.78 (2×)

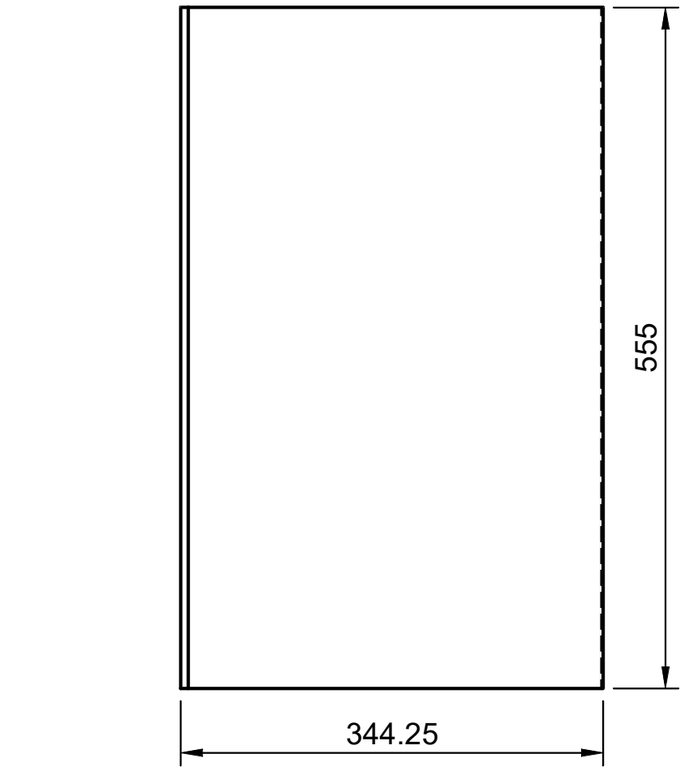
DWG No.

FLEXY-15

Rev.	Date of issue	Sheet
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Dept. SdWa5	Technical reference Scale 1:2 on A3	Created by Oliver Ripper 25.02.2022	Approved by Stefan Ripper
Material **18 mm** **multiplex plywood**		Document type Build plan Title MrFlexySMPs Folded Horn Hybrid **Strip 60 × 555**	Document status revised DWG No. FLEXY-16
			Rev. D / Date of issue 2026-10-17 / Sheet 16/26





Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

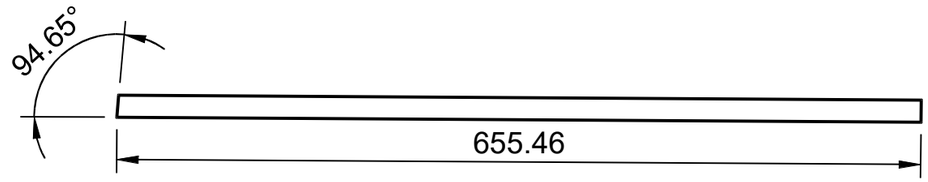
Document status: revised

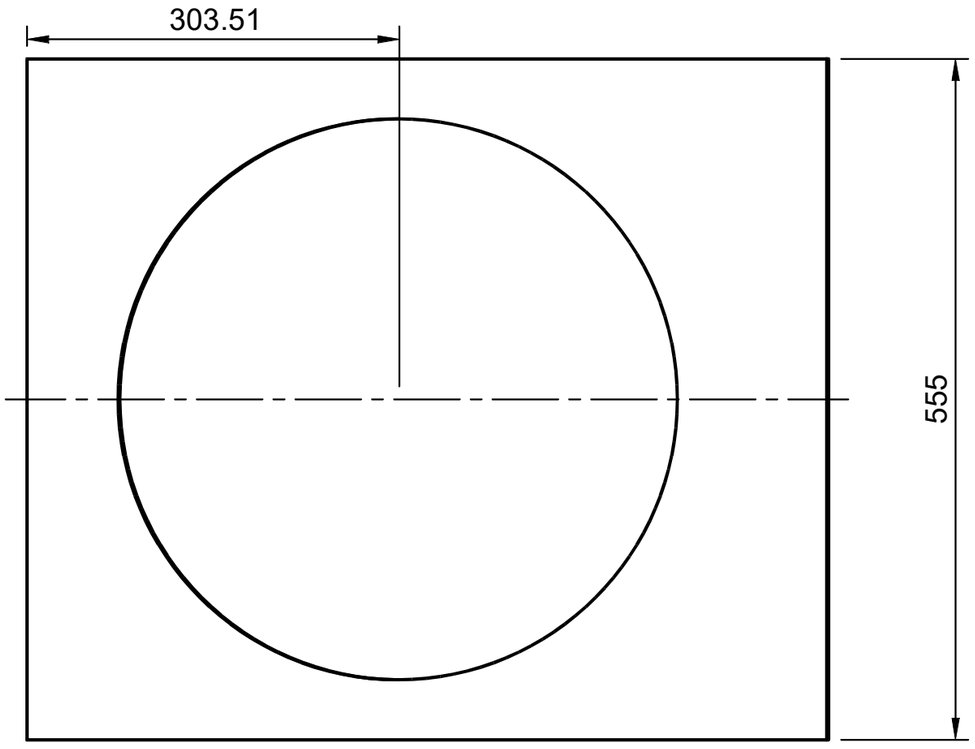
Title: MrFlexySMPs Folded Horn Hybrid

Panel 344.25 × 555

DWG No.: FLEXY-17

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Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

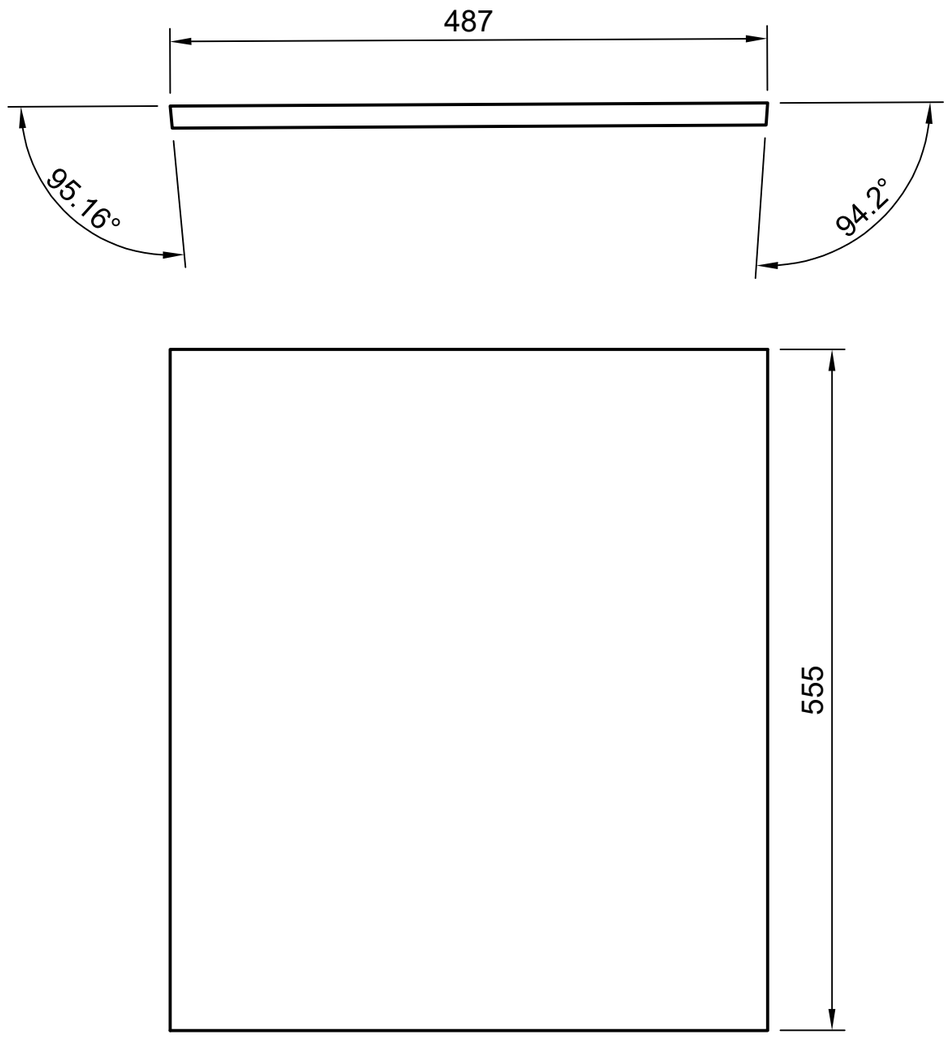
MrFlexySMPs
Folded Horn Hybrid

Baffle with driver cutout

DWG No.

FLEXY-18

Rev.	Date of issue	Sheet
D	2026-10-17	18/26



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

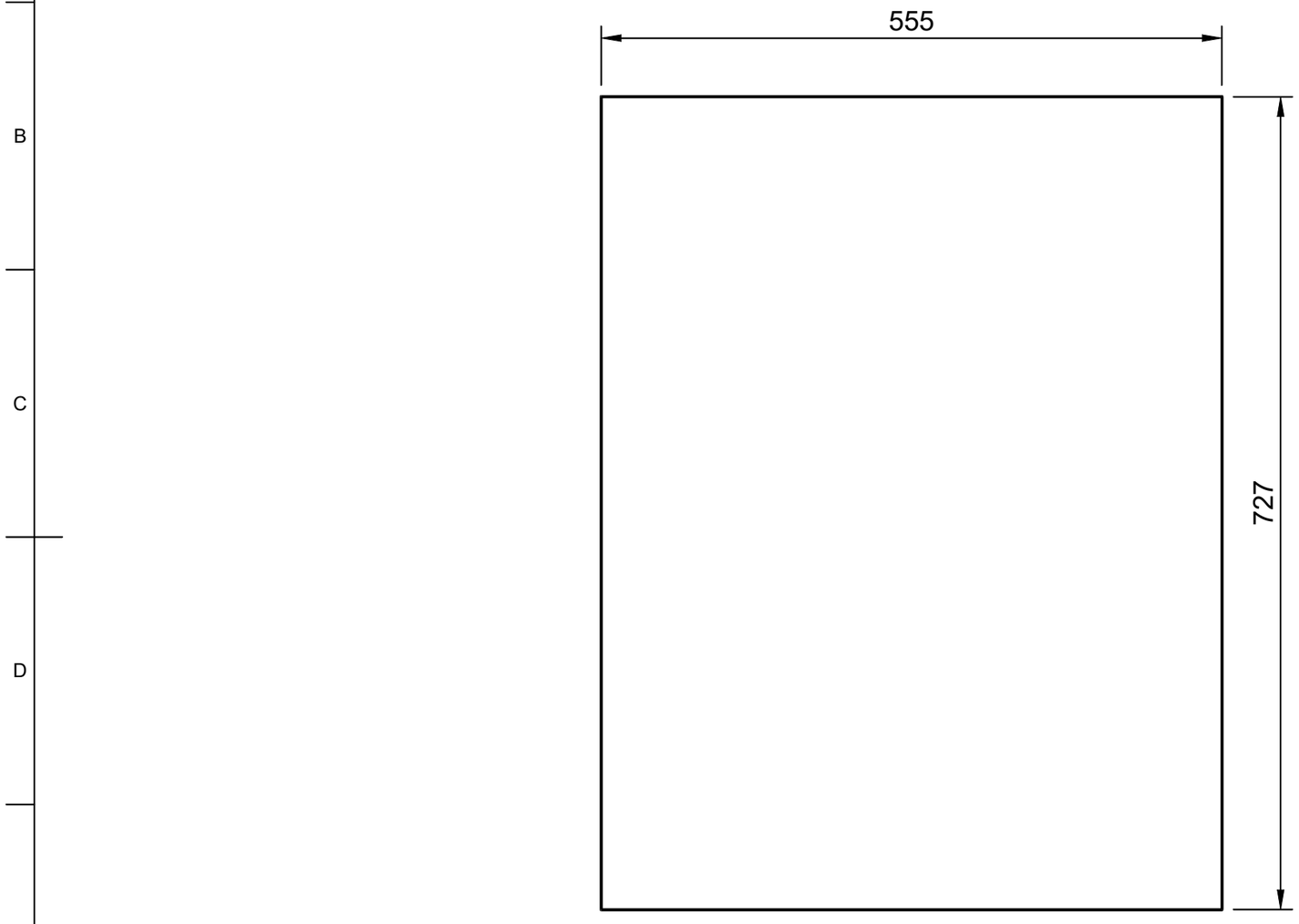
Document status: revised

Title: MrFlexySMPs
Folded Horn Hybrid

Panel 487 × 555

DWG No.: FLEXY-19

Rev.	Date of issue	Sheet
D	2026-10-17	19/26



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

MrFlexySMPs
Folded Horn Hybrid

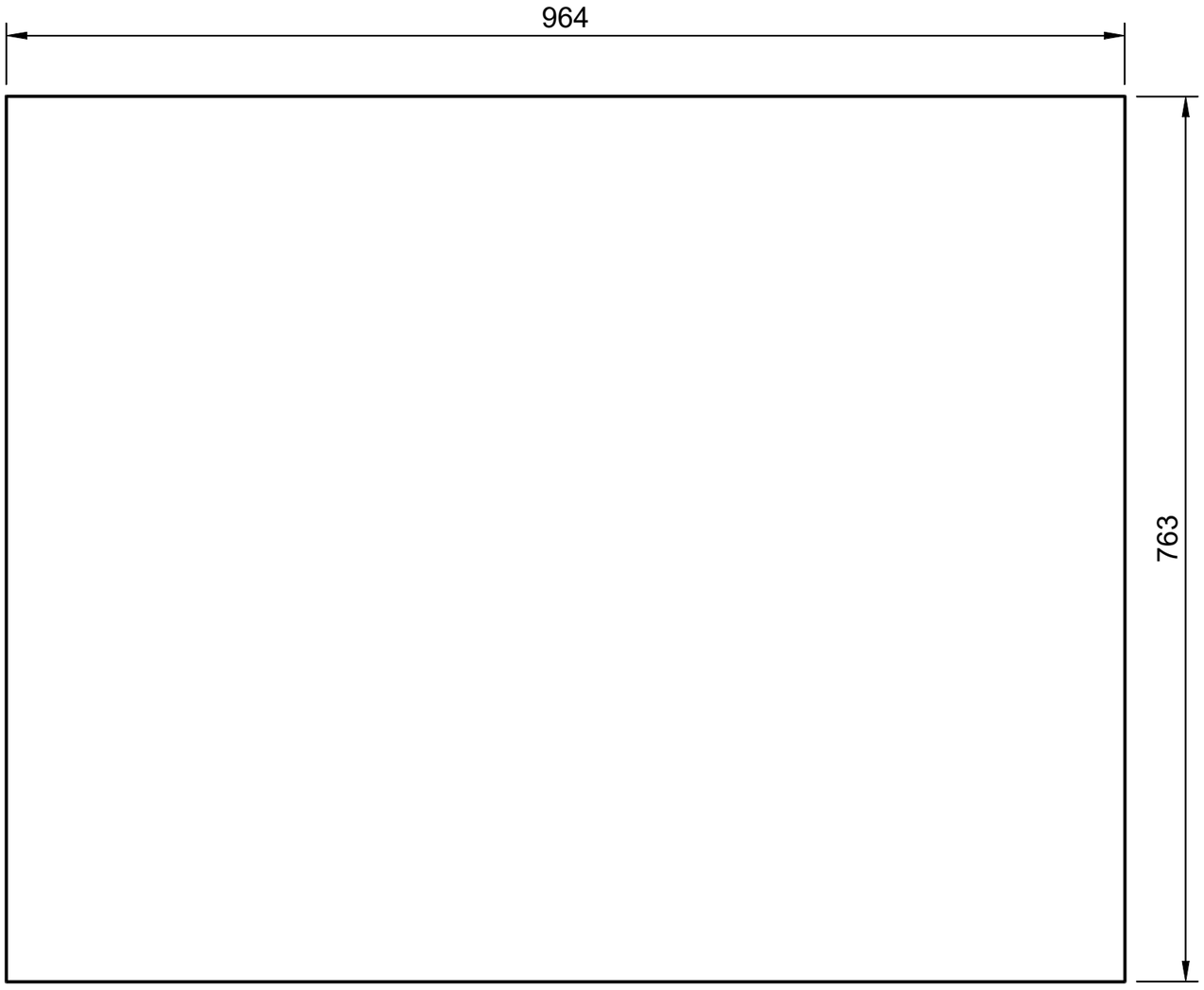
Panel 555 × 727

DWG No.

FLEXY-20

Rev.	Date of issue	Sheet
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x2



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type	Document status
Build plan	revised

Title

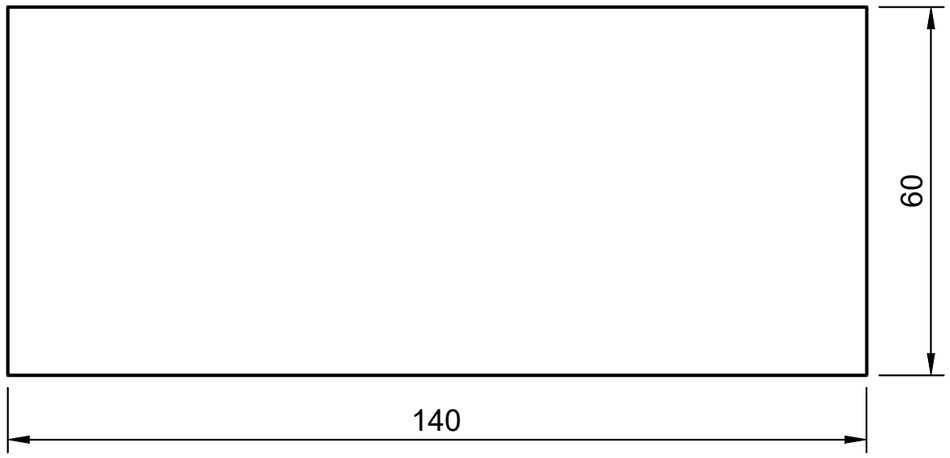
MrFlexySMPs
Folded Horn Hybrid

Side wall (2×)

DWG No.

FLEXY-21

Rev.	Date of issue	Sheet
D	2026-10-17	21/26



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:1 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

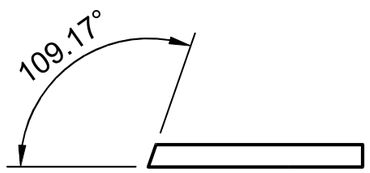
Document status: revised

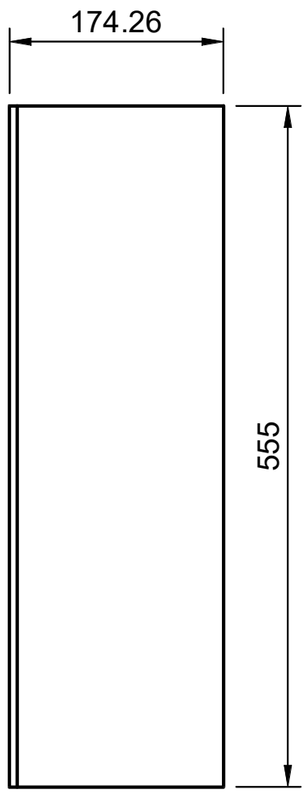
Title: MrFlexySMPs
Folded Horn Hybrid

Block 140 × 60

DWG No.: FLEXY-22

Rev.	Date of issue	Sheet
D	2026-10-17	22/26





Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

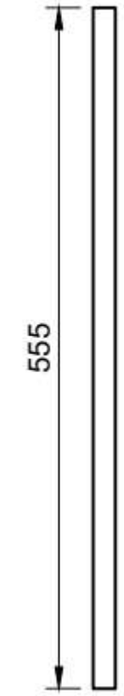
Document status: revised

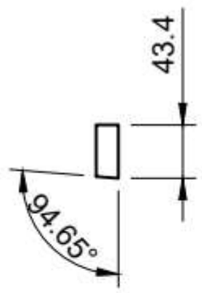
Title: MrFlexySMPs
Folded Horn Hybrid

Panel 174.26 × 555

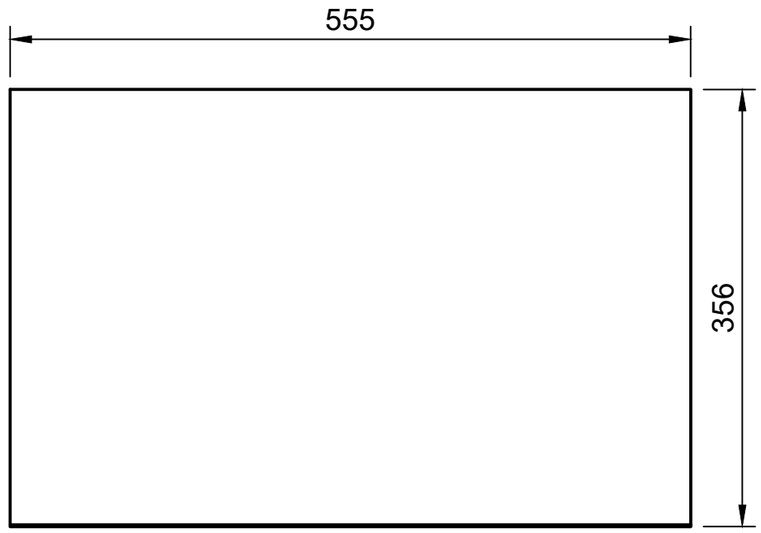
DWG No.: FLEXY-23

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Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper
Material		Document type	Document status
18 mm		Build plan	revised
multiplex plywood		Title MrFlexySMPs Folded Horn Hybrid **Strip 555**	DWG No. FLEXY-24
			Rev. D / Date of issue 2026-10-17 / Sheet 24/26



Dept.	Technical reference	Created by	Approved by
SdWa5	Scale 1:5 on A3	Oliver Ripper 25.02.2022	Stefan Ripper

Material

18 mm

multiplex plywood

Document type: Build plan

Document status: revised

Title: MrFlexySMPs
Folded Horn Hybrid

Panel 555 × 356

DWG No.: FLEXY-25

Rev.	Date of issue	Sheet
D	2026-10-17	25/26

Parts list

Sheet	Part	L × W	Qty	Note
4	Panel	555 × 293.12	1	angle 109.17°
5	Bottom panel	964 × 555	1	same blank as the top panel (sheet 6)
6	Top panel	964 × 555	1	lid opening 510 × 498, the cut-out becomes the lid (sheet 14)
7	Frame (optional)	556 × 555	1	inner 474 × 462, or four battens from offcuts
8	Trapezoid strip	286.74 × 54.68	2	angle 109.17°
9	Brace (optional)	555 × 100	1	
10	Brace (optional)	555 × 127.89	1	angle 98.85°
11	Brace plate (optional)	559.85 × 463	1	contour per sheet
12	Strip	555 × 77.78	1	angle 45°
13	Panel	555 × 296.98	1	angle 45°
14	Lid (optional)	508 × 496	1	cut out of the top panel (sheet 6), no blank of its own
15	Trapezoid	322.78 × 126.67	2	angle 94.65°
16	Strip	555 × 60	1	
17	Panel	555 × 344.25	1	angles 95.16° and 70.83°
18	Baffle	655.46 × 555	1	driver cut-out per sheet, angle 94.65°
19	Panel	555 × 487	1	angles 94.2° and 95.16°
20	Panel	727 × 555	1	
21	Side wall	964 × 763	2	inner contour per sheet
22	Block	140 × 60	1	
23	Panel	555 × 174.26	1	angle 109.17°
24	Strip	555 × 43.4	1	angle 94.65°
25	Panel	555 × 356	1	

All parts 18 mm multiplex plywood. L × W are the outer sizes of the sheets, bevels included. The lid has no blank of its own.

Cut list for cutlistoptimizer.com

Import the CSV, sizes rounded up to whole mm. Optional parts are disabled.

Length	Width	Qty	Material	Label	Enabled
555	294	1	Multiplex 18mm	FLEXY-04 Panel	true
964	555	1	Multiplex 18mm	FLEXY-05 Bottom panel	true
964	555	1	Multiplex 18mm	FLEXY-06 Top panel	true
556	555	1	Multiplex 18mm	FLEXY-07 Frame	false
287	55	2	Multiplex 18mm	FLEXY-08 Trapezoid strip	true
555	100	1	Multiplex 18mm	FLEXY-09 Brace	false
555	128	1	Multiplex 18mm	FLEXY-10 Brace	false
560	463	1	Multiplex 18mm	FLEXY-11 Brace plate	false
555	78	1	Multiplex 18mm	FLEXY-12 Strip	true
555	297	1	Multiplex 18mm	FLEXY-13 Panel	true
323	127	2	Multiplex 18mm	FLEXY-15 Trapezoid	true
555	60	1	Multiplex 18mm	FLEXY-16 Strip	true
555	345	1	Multiplex 18mm	FLEXY-17 Panel	true
656	555	1	Multiplex 18mm	FLEXY-18 Baffle	true
555	487	1	Multiplex 18mm	FLEXY-19 Panel	true
727	555	1	Multiplex 18mm	FLEXY-20 Panel	true
964	763	2	Multiplex 18mm	FLEXY-21 Side wall	true
140	60	1	Multiplex 18mm	FLEXY-22 Block	true
555	175	1	Multiplex 18mm	FLEXY-23 Panel	true
555	44	1	Multiplex 18mm	FLEXY-24 Strip	true
555	356	1	Multiplex 18mm	FLEXY-25 Panel	true

CSV: https://github.com/SdWa5/3d/blob/main/published/flexy-folded-horn-hybrid/flexy-cutlist.csv

Recommended stock sheets, 18 mm

3050 × 1500 mm: at least 2 sheets (2 with the optional parts)
3000 × 1500 mm: at least 2 sheets (2 with the optional parts)
2500 × 1250 mm: at least 2 sheets (2 with the optional parts)

A lower bound from the blank area of 4.62 m² (5.31 m² with the optional parts). Kerf and nesting add to it, so the optimizer gives the real count.

Dept.: SdWa5	Technical reference: No scale, A3	Created by: Oliver Ripper 25.02.2022	Approved by: Stefan Ripper
Material: **18 mm** **multiplex plywood**		Document type: Build plan	Document status: revised
		Title: MrFlexySMPs Folded Horn Hybrid **Parts list and cut list**	DWG No.: FLEXY-26
			Rev.: D · Date of issue: 2026-10-17 · Sheet: 26/26